

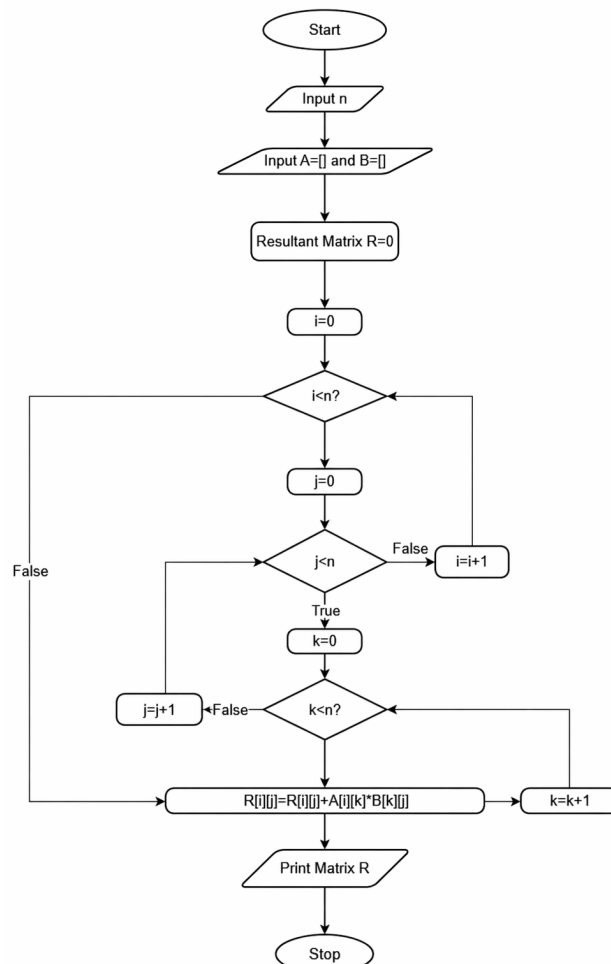
8.1.1. Multiplication of Two Square Matrices.README.md

AIM: Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Algorithm

- 1.Start
- 2.Input the order n of the square matrices.
- 3.Input elements of Matrix A and Matrix B.
- 4.Create a result matrix R of size $n \times n$ and initialize all elements to 0.
- 5.Initialize loop variable i from 0 to $n-1$.
- 6.Inside that, initialize loop variable j from 0 to $n-1$.
- 7.Inside that, initialize loop variable k from 0 to $n-1$.
- 8.Perform multiplication and addition: If inside loops $\rightarrow R[i][j] = R[i][j] + A[i][k] \times B[k][j]$
- 9.Repeat until all loops finish.
- 10.Print the result matrix R.
- 11.Stop

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Execution

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3.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension:" and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

Multiplication...

```
1 n = int(input("dimension: "))
2 print("first matrix:")
3 A = []
4
5 for i in range(n):
6     row = list(map(int, input().split()))
7     A.append(row)
8
9 print("second matrix:")
10 B = []
11
12 for i in range(n):
13     row = list(map(int, input().split()))
14     B.append(row)
15
16 R = []
17 for i in range(n):
18     R.append([0] * n)
19
20 for i in range(n):
```

Average time

0.012 s

12.00 ms

Maximum time

0.014 s

14.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

Actual output

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

Terminal

Test cases

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